High-Order Graph Regularized Sparse Projection for Unsupervised Hyperspectral Band Selection

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***Abstract*—Hyperspectral images (HSIs) usually contain hundreds of highly correlated spectral bands, which may introduce severe redundancy, high computational cost, and the Hughes phenomenon. Band selection (BS) reduces spectral dimensionality by selecting a compact subset of original bands while preserving their physical interpretability. However, existing methods mainly exploit pairwise spectral or spatial dependencies, leaving complex high-order correlations insufficiently characterized. Moreover, manifold preservation and sparse BS are rarely integrated into a unified framework. To address these issues, this article proposes a high-order graph-regularized sparse projection method for unsupervised hyperspectral BS. Based on region-level samples generated by superpixels, the method constructs a high-order graph through multistep neighborhood propagation, thereby alleviating the limitations of traditional first-order graphs in characterizing complex sample relationships and more comprehensively capturing the intrinsic structure of HSI data. Moreover, self-representation learning, graph regularization, and row-sparse projection are integrated into a unified optimization framework. Graph consistency and Laplacian regularization preserve high-order structural relationships and local manifold information, while row sparsity enables explicit estimation of band importance. Consequently, the proposed method selects a compact band subset with strong discriminative capability and representativeness while reducing spectral redundancy. The source code is available at https://github.com/CPREgroup/hogrsp**

***Index Terms*—Band selection (BS), high-order graph, hyperspectral image (HSI), self-representation, sparse projection, unsupervised learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of contiguous spectral bands and provide rich spectral information to distinguish different land-cover classes. Benefiting from their strong spectral discrimination capabilities, HSIs have been widely used in various remote sensing tasks, such as land-cover classification, anomaly detection, target detection, and environmental monitoring [1], [2], [3], [4], [5]. However, the large number of spectral bands also introduces severe spectral redundancy, high computational cost, and the Hughes phenomenon, which may degrade the performance of downstream tasks, especially when only limited labeled samples are available. Therefore, dimensionality reduction is an essential preprocessing step for HSI analysis [6], [7], [8].

Dimensionality reduction methods for HSIs can generally be divided into feature extraction and band selection (BS). Although feature extraction methods such as principal component analysis (PCA), linear discriminant analysis (LDA), manifold-based learning approaches, and spectral–spatial feature extraction methods [9] are capable of projecting raw data onto a low-dimensional feature space, thereby effectively reducing the dimensionality of the data, these transformed features may compromise the physical interpretability of the original spectral bands. On the contrary, BS aims to choose a subset of representative bands directly from the original HSI without spectral transformation. Therefore, the selected bands can preserve the physical meaning and inherent properties of the original data, making BS more suitable for practical HSI applications [10]. Since collecting sufficient labeled HSI samples is usually expensive and time-consuming, efficient unsupervised BS [11] has become a focus of research in this field.

Traditional unsupervised BS methods mainly include searching-based, ranking-based, and clustering-based approaches. They obtain representative band subsets by optimizing predefined objectives [12], [13], evaluating individual band importance [14], [15], or grouping correlated bands [16], [17], [18], [19]. Although effective in reducing spectral redundancy, these methods primarily operate at the band level and often overlook the intrinsic relationships among HSI samples. Recently, graph-based and self-representation-based methods have attracted increasing attention because they can model the relationships among spectral bands or samples through graph learning, subspace representation, and graph regularization [11], [20], [21], [22], [23], [24], [25], [26]. However, most existing methods still rely on pairwise similarities or first-order neighborhood relationships, which may be insufficient to describe complex region-level structures in HSIs. In addition, representative BS is often performed after graph construction or clustering, making the final band subset dependent on the learned affinity structure.

Motivated by the above observations, we propose a high-order graph-regularized sparse projection model for unsupervised hyperspectral BS. The proposed model is designed to improve the characterization of intrinsic relationships among samples and to preserve this information during the learning of the importance of the bands. By connecting high-order structure modeling with sparse projection, it provides a

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structure-aware solution for selecting compact and informative band subsets.

The main contributions of this article are summarized as follows.

1) We propose a region-level high-order graph construction strategy for unsupervised hyperspectral BS. By exploiting multistep neighborhood relationships among homogeneous superpixel-level samples, the constructed graph captures structural dependencies beyond direct pairwise connections and provides a more comprehensive characterization of the intrinsic structure of HSIs.
2) We develop a graph-regularized sparse projection model for BS, which unifies structural preservation and informative BS into a single optimization framework. By incorporating the learned structural prior into the projection process, the proposed model preserves complex relationships among HSI samples while learning a row-sparse projection matrix to explicitly estimate band importance.
3) Extensive experiments on three public HSI datasets demonstrate the effectiveness of the proposed method in terms of classification performance and band subset quality, compared with eight representative unsupervised BS methods.

The remainder of this article is organized as follows. Section II reviews representative unsupervised HSI band selection methods. Section III presents the proposed model, including its construction, optimization strategy, and computational complexity. Section IV describes the experimental setup and reports the comparative experiments, ablation studies, and model analyses. Finally, Section V concludes this article.

II. Related Work

This section reviews representative HSI BS methods that are closely related to this work. According to the selection strategy, existing methods can be roughly divided into searching-based, ranking-based, and clustering-based or graph-based methods.

A. Search-Based BS

Search-based BS methods formulate BS as a combinatorial or constrained optimization problem and search for an optimal or suboptimal band subset by optimizing predefined objective functions [27]. Early methods usually adopt greedy strategies to iteratively add or remove bands from the original band set. For example, volume gradient-based BS treats each band as a point and removes highly correlated bands by measuring the volume variation of the constructed parallelotope [28]. To enhance the global search capability, evolutionary and swarm intelligence algorithms have also been introduced into HSI BS. Representative methods include clonal selection-based BS [29], artificial bee colony-based methods [30], gray wolf optimizer-based methods [31], and multiobjective evolutionary BS methods [12], [32]. Recently, multiobjective unsupervised BS and heterogeneous cuckoo search have also been used to balance band information, redundancy, and global exploration capability [13], [33].

Benefiting from heuristic or evolutionary search strategies, searching-based methods can explore a large solution space and often obtain effective band subsets [34], [35], [36]. However, they usually suffer from relatively high computational complexity when the number of candidate bands is large. In addition, their performance can be sensitive to the design of objective functions, population initialization, and algorithm-specific parameters.

B. Ranking-Based BS

Ranking-based BS methods first evaluate the importance of each band according to a predefined criterion and then select the top-ranked bands as the final subset [37]. The key issue of this type of method is to design an effective band evaluation criterion. Typical criteria include variance, information entropy, mutual information, information divergence, and band dissimilarity. For example, maximum-variance PCA selects bands with large variance by analyzing the covariance matrix of hyperspectral data [38]. Information entropy feature preservation-based BS constructs an entropy feature curve and selects representative bands according to information variation [39]. Mutual information-based methods measure the statistical dependence between bands and select bands with high information content [14]. More recently, an unsupervised band-selection method based on spectral–spatial out-of-distribution information was developed to rank informative bands for hyperspectral anomaly detection [40]. Constrained BS introduces linear constraints to select representative bands that are less affected by interference [15].

Ranking-based methods are simple, efficient, and easy to implement because they usually avoid complex combinatorial optimization. Nevertheless, most of them evaluate each band independently or mainly rely on predefined statistical criteria. As a result, the correlations and redundancy among selected bands may not be fully considered, and several highly similar neighboring bands may be simultaneously selected. Moreover, simple ranking criteria are often insufficient to characterize the complex structure of HSI data.

C. Clustering-Based and Graph-Based BS

Clustering-based BS methods group spectral bands according to their similarities and select representative bands from different clusters to form the final subset. By assigning highly correlated bands to the same cluster, these methods can effectively reduce redundancy among the selected bands. Early studies introduced hierarchical clustering or k-means clustering for hyperspectral band partition [41]. Later, adaptive subspace partition strategies were developed to divide spectral bands into more coherent spectral subspaces according to band similarity [18]. Some methods further introduced determinantal point process models, optimal clustering frameworks, or subspace clustering models to improve band grouping and representative BS [16], [17], [20].

Recently, graph-based and region-aware clustering methods have attracted increasing attention. These methods exploit the spectral–spatial structure of HSIs for more reliable BS. For example, regionwise multiple graph fusion constructs

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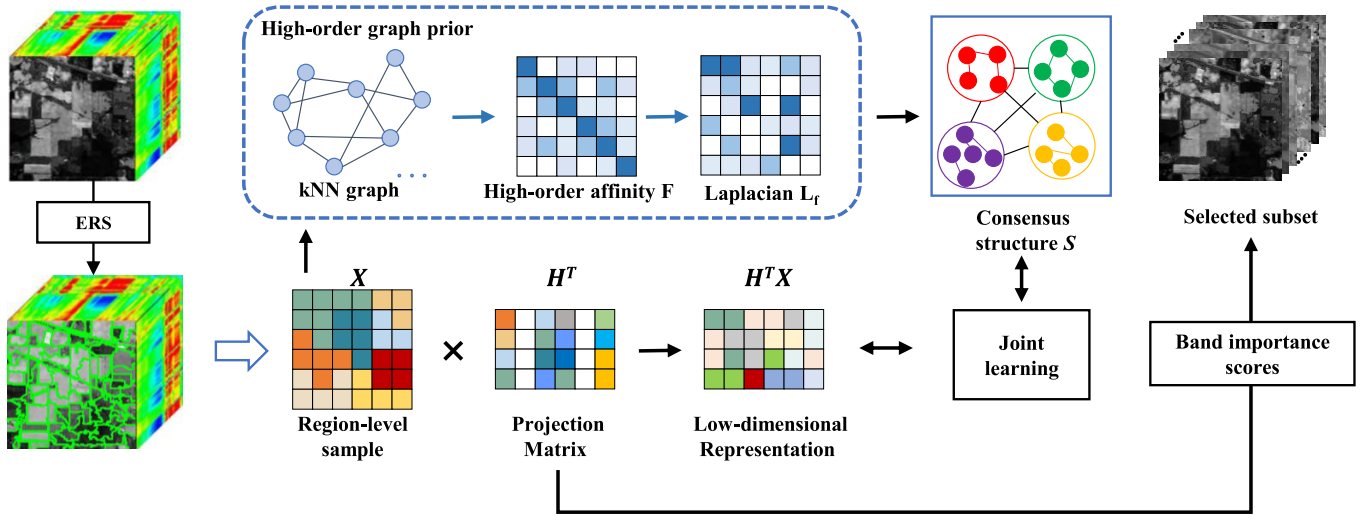


Fig. 1. Overall framework of the proposed method. The original hyperspectral cube is first transformed into superpixel-level region samples. A high-order graph prior is constructed from these region-level samples, where the high-order affinity matrix **F** guides the learning of the structure matrix **S** and the graph Laplacian $\mathbf{L}_f$ preserves the neighborhood structure in the projected subspace. The projection matrix **H** and the structure matrix **S** are jointly optimized, and the final band subset is obtained according to the row norms of **H**.

several similarity graphs from segmented regions and fuses them for spectral clustering [23]. Region-aware latent feature fusion clustering learns latent representations from different region-aware band features and then performs clustering to identify representative bands [42]. Tensorial global–local graph self-representation further explores global and local graph structures and imposes a tensor low-rank constraint on self-representation coefficients to capture high-order band correlations [26].

Although clustering-based and graph-based methods have achieved promising performance, their results often depend on the quality of band partition or the learned affinity structure. Moreover, many existing methods mainly focus on band-to-band similarity, while sample-level structural relationships are not sufficiently preserved during BS. Most graph-based methods also rely on pairwise similarity graphs and mainly exploit first-order neighborhood information, which may be insufficient to describe complex high-order relationships among region-level samples.

In addition to the above methods, deep learning-based BS methods have also been developed in recent years. Some studies employ convolutional neural networks, autoencoders, attention mechanisms, or reinforcement learning to learn band importance in an end-to-end manner [43], [44], [45], [46], [47], [48], [49]. Although deep models can achieve competitive performance, their architectures, loss functions, and training strategies usually require careful design, and the learned selection mechanism is less interpretable. In contrast to these methods, this article focuses on an explicit unsupervised BS model, where sparse projection learning is guided by region-level high-order graph regularization.

III. Methodology

This section details the proposed high-order graph-regularized sparse projection model for unsupervised hyperspectral BS. As presented in Fig. 1, our method comprises four core modules: region-level sample construction, high-order graph prior modeling, joint self-representation and projection matrix learning, and band ranking. Each module is introduced in Sections III-A–III-C, respectively.

A. Model Construction

Let $\mathcal{X} \in \mathbb{R}^{H_0 \times W_0 \times d}$ denote the original hyperspectral cube, where H_0 and W_0 are the spatial height and width, and d is the number of spectral bands. Directly using all pixels as samples usually leads to high computational cost and may introduce redundant local details. Therefore, we first construct a compact region-level representation by superpixel segmentation. In particular, the first principal component of $\mathcal{X}$ is segmented into a set of homogeneous regions by entropy-rate superpixel segmentation [50]. The spectral feature of each region is then obtained by averaging all hyperspectral pixels within the same superpixel. In this way, the original HSI is transformed into a region-level sample matrix

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{d \times n} \tag{1}$$

where n is the number of superpixel-level samples and each column $\mathbf{x}_i \in \mathbb{R}^d$ denotes the mean spectral vector of the ith superpixel region.

To learn a low-dimensional representation for BS, we introduce a projection matrix $\mathbf{H} \in \mathbb{R}^{d \times k}$, where $k \ll d$ is the latent subspace dimension. The projected region-level data are defined as follows:

$$\mathbf{Y} = \mathbf{H}^\top \mathbf{X} \tag{2}$$

where $\mathbf{Y} \in \mathbb{R}^{k \times n}$. Since the ith row of **H** corresponds to the ith spectral band, the rowwise sparsity of **H** can be used to measure band importance after optimization.

Following the self-representation assumption, each projected sample can be reconstructed by other projected samples.

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Thus, the basic self-representation model is formulated as follows:

$$\min_{\mathbf{H},\mathbf{S}} \left\|\mathbf{H}^\top\mathbf{X} - \mathbf{H}^\top\mathbf{X}\mathbf{S}\right\|_F^2 \tag{3}$$

where $\mathbf{S} \in \mathbb{R}^{n\times n}$ is the self-representation matrix among region-level samples. To avoid the trivial solution that each sample is represented by itself, the diagonal entries of $\mathbf{S}$ are constrained to be zero.

Although the self-representation term captures sample relationships in the projected subspace, it may be insufficient to characterize complex correlations among region-level samples. To address this limitation, we construct a high-order graph prior from the original region-level features. In particular, a K-nearest-neighbor (KNN) graph is constructed to obtain the neighborhood set $\mathcal{N}_i$ of each sample. The initial affinity matrix is defined as follows:

$$C_{ij} = \begin{cases} \exp\left(-\dfrac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|_2^2}{2\sigma^2}\right), & \mathbf{x}_j \in \mathcal{N}_i \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where σ denotes the Gaussian kernel bandwidth. The resulting affinity matrix is symmetrized as $\mathbf{C} = (\mathbf{C} + \mathbf{C}^\top)/2$. Before propagation, $\mathbf{C}$ is normalized rowwise.

Starting from the first-order affinity matrix $\mathbf{C}$, the vth-order propagation matrix is recursively computed as follows:

$$\mathbf{C}^1 = \mathbf{C}, \quad \mathbf{C}^v = \mathbf{C}^{v-1}\mathbf{C}, \quad v = 2, \ldots, V \tag{5}$$

where the products denote standard matrix multiplication. Thus, $\mathbf{C}^v$ captures affinities propagated through v successive neighborhood connections in the original KNN graph.

The high-order graph prior is obtained by aggregating the affinities from different propagation orders.

$$\mathbf{F} = \sum_{v=1}^{V} \mathbf{C}^v \tag{6}$$

where V denotes the maximum propagation order. The first-order term retains direct neighborhood relationships, whereas the higher order terms capture indirect dependencies through intermediate samples. After aggregation, $\mathbf{F}$ is symmetrized, its diagonal entries are set to zero, and it is normalized by its maximum element. Consequently, $\mathbf{F}$ retains local neighborhood information while incorporating structural dependencies beyond immediate neighbors.

As shown in Fig. 1, the high-order graph prior is incorporated into the proposed model from two aspects. First, the high-order affinity matrix $\mathbf{F}$ is used to guide the learned self-representation matrix $\mathbf{S}$

$$\min_{\mathbf{S}} \|\mathbf{S} - \mathbf{F}\|_F^2 . \tag{7}$$

This term encourages the learned structure matrix to be consistent with the high-order region-level relationships encoded by $\mathbf{F}$.

Second, the Laplacian matrix associated with $\mathbf{F}$ is used to preserve high-order neighborhood relationships in the projected subspace. Let $\mathbf{L}_f = \mathbf{D}_f - \mathbf{F}$, where $\mathbf{D}_f$ is the diagonal degree matrix with $(\mathbf{D}_f)_{ii} = \sum_{j=1}^{n} F_{ij}$. The high-order Laplacian regularization is formulated as follows:

$$\min_{\mathbf{H}} \operatorname{Tr}\left(\mathbf{H}^\top\mathbf{X}\mathbf{L}_f\mathbf{X}^\top\mathbf{H}\right). \tag{8}$$

Since $\mathbf{Y} = \mathbf{H}^\top\mathbf{X}$, the regularization term in (8) can be equivalently expressed as follows:

$$\begin{aligned}\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{X}\mathbf{L}_f\mathbf{X}^\top\mathbf{H}\right) &= \operatorname{Tr}\left(\mathbf{Y}\mathbf{L}_f\mathbf{Y}^\top\right) \\ &= \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} F_{ij}\left\|\mathbf{y}_i - \mathbf{y}_j\right\|_2^2 .\end{aligned} \tag{9}$$

where $\mathbf{y}_i = \mathbf{H}^\top\mathbf{x}_i$ denotes the projected representation of the ith region-level sample. Therefore, samples with strong high-order affinities are encouraged to remain close after projection.

To obtain a band-discriminative projection matrix, the $\ell_{2,1}$-norm is imposed on $\mathbf{H}$ to induce rowwise sparsity, where $\|\mathbf{H}\|_{2,1} = \sum_{i=1}^{d}\|\mathbf{H}_{i,:}\|_2$. Since each row corresponds to one original spectral band, less informative bands tend to be suppressed by shrinking their row norms. Thus, $\|\mathbf{H}_{i,:}\|_2$ can be directly used as the importance score of the ith band.

In addition, to stabilize sparse projection learning and reduce redundancy among projection directions, a covariance-aware soft orthogonal regularization is introduced

$$\min_{\mathbf{H}} \left\|\mathbf{H}^\top\mathbf{G}\mathbf{H} - \mathbf{I}_k\right\|_F^2 \tag{10}$$

where $\mathbf{G} = 1/n\mathbf{X}\mathbf{X}^\top + \epsilon\mathbf{I}_d$ is the regularized data covariance matrix and $\epsilon > 0$ is a small constant for numerical stability. This term encourages the learned projection directions to be approximately orthogonal under the regularized data covariance metric.

By combining the self-representation term, high-order graph guidance, Laplacian structure preservation, rowwise sparse projection, and soft orthogonal regularization, the final objective function is formulated as follows:

$$\begin{aligned}\min_{\mathbf{H},\mathbf{S}} \quad & \left\|\mathbf{H}^\top\mathbf{X} - \mathbf{H}^\top\mathbf{X}\mathbf{S}\right\|_F^2 + \alpha\|\mathbf{H}\|_{2,1} + \gamma\|\mathbf{S} - \mathbf{F}\|_F^2 \\ & + \eta\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{X}\mathbf{L}_f\mathbf{X}^\top\mathbf{H}\right) + \lambda\left\|\mathbf{H}^\top\mathbf{G}\mathbf{H} - \mathbf{I}_k\right\|_F^2 \\ \text{s.t.} \quad & \mathbf{S} \geq 0, \quad \operatorname{diag}(\mathbf{S}) = \mathbf{0}\end{aligned} \tag{11}$$

where α, γ, η, and λ are the tradeoff parameters. In (11), $\|\mathbf{H}^\top\mathbf{X} - \mathbf{H}^\top\mathbf{X}\mathbf{S}\|_F^2$ preserves the self-representation structure in the projected subspace, $\|\mathbf{H}\|_{2,1}$ promotes band-level sparsity, $\|\mathbf{S} - \mathbf{F}\|_F^2$ guides $\mathbf{S}$ with the high-order graph prior, $\operatorname{Tr}(\mathbf{H}^\top\mathbf{X}\mathbf{L}_f\mathbf{X}^\top\mathbf{H})$ preserves high-order neighborhood relationships, and $\|\mathbf{H}^\top\mathbf{G}\mathbf{H} - \mathbf{I}_k\|_F^2$ improves the stability of projection learning. After optimization, the bands are ranked according to the row norms of $\mathbf{H}$, and the top-ranked bands are selected as the final band subset.

B. Optimization Strategy

The objective function in (11) involves two coupled variables, i.e., the projection matrix $\mathbf{H}$ and the self-representation matrix $\mathbf{S}$. Since it is difficult to optimize them simultaneously, an alternating optimization strategy is adopted. In each iteration, one variable is updated while the other one is fixed. Let t denote the current iteration index, and let $T_{\max}$ denote the maximum number of alternating optimization iterations. The

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iterations terminate when t reaches $T_{\max}$. The high-order graph prior $\mathbf{F}$ and its Laplacian matrix $\mathbf{L}_f$ are precomputed and fixed during optimization.

The regularized covariance matrix $\mathbf{G}$ is defined in the soft orthogonal regularization term. The projection matrix $\mathbf{H}$ is initialized by PCA, while $\mathbf{S}$ is initialized by $\mathbf{F}$ after removing the diagonal elements.

1) Update of S: With $\mathbf{H}$ fixed, let $\mathbf{Y} = \mathbf{H}^\top\mathbf{X}$. The subproblem with respect to $\mathbf{S}$ is

$$\begin{aligned}\min_{\mathbf{S}} \quad & \|\mathbf{Y}-\mathbf{Y}\mathbf{S}\|_F^2+\gamma\|\mathbf{S}-\mathbf{F}\|_F^2\\ \text{s.t.} \quad & \mathbf{S}\geq 0, \quad \operatorname{diag}(\mathbf{S})=\mathbf{0}.\end{aligned} \tag{12}$$

The zero-diagonal constraint avoids the trivial representation of each sample by itself.

Since the objective in (12) is separable with respect to the columns of $\mathbf{S}$, $\mathbf{S}$ can be updated column by column. Denote the jth columns of $\mathbf{Y}$, $\mathbf{S}$, and $\mathbf{F}$ as $\mathbf{y}_j$, $\mathbf{s}_j$, and $\mathbf{f}_j$, respectively. To satisfy $s_{jj}=0$, the jth coefficient is removed from the optimization variable. Let $\mathcal{I}_j=\{1,2,\ldots,n\}\setminus\{j\}$, $\tilde{\mathbf{Y}}_j=\mathbf{Y}_{\mathcal{I}_j}$, and $\tilde{\mathbf{f}}_j=\mathbf{f}_j(\mathcal{I}_j)$. Then, the jth column subproblem can be written as follows:

$$\begin{aligned}\min_{\mathbf{u}_j} \quad & \left\|\mathbf{y}_j-\tilde{\mathbf{Y}}_j\mathbf{u}_j\right\|_2^2+\gamma\left\|\mathbf{u}_j-\tilde{\mathbf{f}}_j\right\|_2^2\\ \text{s.t.} \quad & \mathbf{u}_j\geq 0\end{aligned} \tag{13}$$

where $\mathbf{u}_j\in\mathbb{R}^{n-1}$ contains the nondiagonal coefficients of $\mathbf{s}_j$.

By expanding (13), it can be equivalently transformed into the following nonnegative quadratic programming problem:

$$\begin{aligned}\min_{\mathbf{u}_j} \quad & \mathbf{u}_j^\top\mathbf{R}_j\mathbf{u}_j-2\mathbf{q}_j^\top\mathbf{u}_j\\ \text{s.t.} \quad & \mathbf{u}_j\geq 0\end{aligned} \tag{14}$$

where $\mathbf{R}_j=\tilde{\mathbf{Y}}_j^\top\tilde{\mathbf{Y}}_j+\gamma\mathbf{I}_{n-1}$ and $\mathbf{q}_j=\tilde{\mathbf{Y}}_j^\top\mathbf{y}_j+\gamma\tilde{\mathbf{f}}_j$. Since $\mathbf{R}_j$ is positive definite when $\gamma>0$, each column subproblem is convex and can be solved by standard nonnegative quadratic programming solvers.

After obtaining the optimal solution $\mathbf{u}_j^\star$, the full column $\mathbf{s}_j$ is reconstructed by the following equation:

$$\mathbf{s}_j\left(\mathcal{I}_j\right)=\mathbf{u}_j^\star, \quad s_{jj}=0. \tag{15}$$

After all columns are updated, the self-representation matrix $\mathbf{S}$ is obtained.

2) Update of H: When $\mathbf{S}$ is fixed, the subproblem with respect to $\mathbf{H}$ is written as follows:

$$\begin{aligned}\min_{\mathbf{H}} \quad & \left\|\mathbf{H}^\top\mathbf{X}-\mathbf{H}^\top\mathbf{X}\mathbf{S}\right\|_F^2+\eta\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{X}\mathbf{L}_f\mathbf{X}^\top\mathbf{H}\right)\\ & +\alpha\|\mathbf{H}\|_{2,1}+\lambda\left\|\mathbf{H}^\top\mathbf{G}\mathbf{H}-\mathbf{I}_k\right\|_F^2.\end{aligned} \tag{16}$$

To simplify the projected self-representation term, let $\mathbf{M}=(\mathbf{I}_n-\mathbf{S})(\mathbf{I}_n-\mathbf{S})^\top$. Then, the first term can be rewritten as follows:

$$\begin{aligned}\left\|\mathbf{H}^\top\mathbf{X}-\mathbf{H}^\top\mathbf{X}\mathbf{S}\right\|_F^2&=\left\|\mathbf{H}^\top\mathbf{X}\left(\mathbf{I}_n-\mathbf{S}\right)\right\|_F^2\\ &=\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{X}\mathbf{M}\mathbf{X}^\top\mathbf{H}\right).\end{aligned} \tag{17}$$

The $\ell_{2,1}$-norm regularization is nonsmooth because it is composed of rowwise Euclidean norms. To handle this term, an iterative reweighted strategy is adopted [51]. Given the current solution $\mathbf{H}^{(t)}$, the diagonal weighting matrix $\mathbf{D}_H^{(t)}$ is defined as follows:

$$\mathbf{D}_H^{(t)}(i,i)=\frac{1}{2\sqrt{\left\|\mathbf{H}_{i,:}^{(t)}\right\|_2^2+\varepsilon}}, \quad i=1,2,\ldots,d \tag{18}$$

where ε is a small positive constant for numerical stability. With $\mathbf{D}_H^{(t)}$ fixed, the $\ell_{2,1}$-norm term is locally approximated by the following reweighted quadratic form:

$$\begin{aligned}\|\mathbf{H}\|_{2,1}&\approx\sum_{i=1}^{d}\mathbf{D}_H^{(t)}(i,i)\left\|\mathbf{H}_{i,:}\right\|_2^2+\kappa^{(t)}\\ &=\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{D}_H^{(t)}\mathbf{H}\right)+\kappa^{(t)}\end{aligned} \tag{19}$$

where $\kappa^{(t)}\in\mathbb{R}$ denotes an iteration-dependent additive constant determined by the fixed current solution $\mathbf{H}^{(t)}$. It is independent of the optimization variable $\mathbf{H}$ within the tth subproblem. Therefore, $\kappa^{(t)}$ can be omitted from the subsequent $\mathbf{H}$ subproblem without affecting its minimizer.

By combining the quadratic terms, the H subproblem at the tth iteration is approximated by the following quadratic surrogate problem:

$$\min_{\mathbf{H}}\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{Q}^{(t)}\mathbf{H}\right)+\lambda\left\|\mathbf{H}^\top\mathbf{G}\mathbf{H}-\mathbf{I}_k\right\|_F^2 \tag{20}$$

where $\mathbf{Q}^{(t)}=\mathbf{X}\mathbf{M}\mathbf{X}^\top+\eta\mathbf{X}\mathbf{L}_f\mathbf{X}^\top+\alpha\mathbf{D}_H^{(t)}$. To solve (20), we use the generalized spectral structure of the matrix pair $(\mathbf{Q}^{(t)},\mathbf{G})$. Since $\mathbf{G}$ is positive definite, we solve the following generalized eigenvalue problem:

$$\mathbf{Q}^{(t)}\mathbf{h}_i=\mu_i\mathbf{G}\mathbf{h}_i, \quad \mathbf{h}_i^\top\mathbf{G}\mathbf{h}_j=\delta_{ij} \tag{21}$$

where μ_i is the generalized eigenvalue, $\mathbf{h}_i$ is the corresponding generalized eigenvector, and δ_{ij} denotes the Kronecker delta. Under this normalization, the generalized eigenvectors form a $\mathbf{G}$-orthonormal basis, which decouples the quadratic surrogate term and the soft orthogonal regularization along different projection directions.

In particular, we represent the projection matrix as follows:

$$\mathbf{H}=[u_1\mathbf{h}_1,u_2\mathbf{h}_2,\ldots,u_k\mathbf{h}_k] \tag{22}$$

where u_i is the adaptive scaling coefficient of the ith projection direction. Substituting (22) into the first term of (20), and using $\mathbf{Q}^{(t)}\mathbf{h}_i=\mu_i\mathbf{G}\mathbf{h}_i$ and $\mathbf{h}_i^\top\mathbf{G}\mathbf{h}_j=\delta_{ij}$, we obtain

$$\operatorname{Tr}\left(\mathbf{H}^\top\mathbf{Q}^{(t)}\mathbf{H}\right)=\sum_{i=1}^{k}\mu_iu_i^2. \tag{23}$$

Similarly, the soft orthogonal regularization term becomes

$$\left\|\mathbf{H}^\top\mathbf{G}\mathbf{H}-\mathbf{I}_k\right\|_F^2=\sum_{i=1}^{k}\left(u_i^2-1\right)^2. \tag{24}$$

Therefore, the problem in (20) is reduced to a set of independent scalar subproblems

$$\min_{u_i}\quad \mu_iu_i^2+\lambda\left(u_i^2-1\right)^2, \quad i=1,2,\ldots,k. \tag{25}$$

The first term measures the contribution of the ith projection direction to the quadratic surrogate, while the second term penalizes the deviation from the desired $\mathbf{G}$-orthonormal scale.

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By setting the derivative of (25) with respect to u_i to zero and comparing the resulting stationary points, the nonnegative optimal scaling coefficient is obtained as follows:

$$u_i = \sqrt{\max\left(1 - \frac{\mu_i}{2\lambda}, 0\right)}, \quad i = 1, 2, \ldots, k. \tag{26}$$

The $\max(\cdot, 0)$ operation forces u_i to be real and nonnegative.

Since $\mathbf{Q}^{(t)}$ is positive semidefinite and $\mathbf{G}$ is positive definite, all generalized eigenvalues are nonnegative. We arrange them as $\mu_1 \leq \mu_2 \leq , \ldots, \leq \mu_d$. After substituting (26) into (25), the minimized objective associated with μ_i is

$$\phi(\mu_i) = \begin{cases} \mu_i - \dfrac{\mu_i^2}{4\lambda}, & 0 \leq \mu_i < 2\lambda \\ \lambda, & \mu_i \geq 2\lambda. \end{cases} \tag{27}$$

Since $\phi(\mu_i)$ is nondecreasing with respect to μ_i, the generalized eigenvectors corresponding to the smallest k generalized eigenvalues are selected as the projection directions.

Finally, the projection matrix is updated by the following equation:

$$\mathbf{H}^{(t+1)} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_k] \operatorname{diag}(u_1, u_2, \ldots, u_k). \tag{28}$$

The two variables are updated alternately until the maximum number of iterations is reached. After the final iteration, the importance of each band is measured by the ℓ_2-norm of the corresponding row of $\mathbf{H}$.

Algorithm 1 Proposed Method

Input: HSI cube $\mathcal{X}$, the number of selected bands K, the maximum propagation order V, trade-off parameters α, γ, η, and λ, and the maximum number of optimization iterations $T_{\max}$.
Output: The obtained band subset.
Segment the HSI by PCA and ERS, and construct the region-level sample matrix $\mathbf{X}$.
Construct the high-order graph $\mathbf{F}$ and its Laplacian matrix $\mathbf{L}_f$.
Initialize $\mathbf{H}$ by PCA and initialize $\mathbf{S}$ by $\mathbf{F}$.
Set $t = 0$.
while $t < T_{\max}$ **do**
    Compute the projected data $\mathbf{Y} = \mathbf{H}^\top \mathbf{X}$.
    Update $\mathbf{S}$ by solving Eq. (14) column by column.
    Compute $\mathbf{M} = (\mathbf{I}_n - \mathbf{S})(\mathbf{I}_n - \mathbf{S})^\top$.
    Update $\mathbf{D}_H^{(t)}$ by Eq. (18) and construct $\mathbf{Q}^{(t)}$.
    Solve Eq. (21) and compute the scaling coefficients by Eq. (26).
    Update $\mathbf{H}$ by Eq. (28).
    Set $t = t + 1$.
end while
Obtain the band subset from the row-norm scores of $\mathbf{H}$.
Return the band subset.

C. Algorithm and Computational Complexity

The overall procedure of the proposed method is summarized in Algorithm 1. The proposed method first converts the original HSI into region-level samples and constructs the high-order graph prior. Then, the self-representation matrix $\mathbf{S}$ and the projection matrix $\mathbf{H}$ are alternately updated until the maximum number of iterations is reached. Finally, the band subset is obtained according to the row-norm scores of $\mathbf{H}$.

Let $p = H_0 W_0$ denote the number of pixels. The preprocessing stage mainly includes superpixel segmentation, region-level sample construction, and high-order graph construction. Its computational complexity is approximately $O(p \log p + pd + n^2 d + Vn^3)$, where $O(p \log p + pd)$ is used for region-level sample construction, $O(n^2 d)$ is used for pairwise similarity calculation, and $O(Vn^3)$ is used for high-order graph propagation.

During the optimization stage, the main cost comes from updating $\mathbf{S}$ and $\mathbf{H}$. Updating $\mathbf{S}$ involves operations on the $n \times n$ self-representation matrix while updating $\mathbf{H}$ requires constructing the quadratic matrix and solving the generalized eigenvalue problem in (21). Therefore, the overall optimization complexity can be roughly estimated as $O(T_{\max}(n^3 + dn^2 + d^2 n + d^3))$. Since the graph prior is constructed on region-level samples rather than all pixels, and usually $n \ll p$, the proposed method avoids the high cost of pixel-level graph modeling while preserving useful structural information for BS.

TABLE I
STATISTICS OF THE DATASETS USED IN OUR EXPERIMENTS

Dataset	Sensor	Size	Bands	Classes	Resolution
IP	AVIRIS	145×145	200	16	20 m
PU	ROSIS	610×340	103	9	1.3 m
SA	AVIRIS	512×217	204	16	3.7 m

IV. EXPERIMENTS

A. Experimental Setup

To evaluate the effectiveness of the proposed method, experiments were conducted on three widely used HSI datasets, including Indian Pines (IPs), Pavia University (PU), and Salinas (SAs). These datasets cover different imaging sensors, spatial resolutions, scene types, and spectral characteristics, providing a comprehensive benchmark for evaluating the robustness and generalization ability of different BS methods. The basic statistics of the three datasets are summarized in Table I.

The IPs dataset was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over an agricultural area in Northwestern Indiana, USA. It contains 145 × 145 pixels with a spatial resolution of 20 m. After removing 24 water-absorption and noisy bands from the original 224 bands, 200 bands are retained for experiments. The corresponding ground truth contains 16 land-cover classes.

The PU dataset was collected by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia, northern Italy. It consists of 610 × 340 pixels with a spatial resolution of 1.3 m. After removing 12 noisy bands from the original 115 bands, 103 bands are used for evaluation. The ground truth contains nine urban land-cover classes.

The SAs dataset was also captured by the AVIRIS sensor over the SAs Valley in California, USA. It has a spatial size

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of 512 × 217 pixels and a spatial resolution of 3.7 m. After discarding 20 water absorption bands from the original 224 bands, 204 bands are retained. The ground truth includes 16 land-cover classes.

To verify the superiority of the proposed method, several representative unsupervised HSI BS methods were selected for comparison, including BS-Net-Conv [45], DARecNet-BS [52], DEHF [53], Dropout-CAE [54], DVS3C [55], HLFC [56], SSANet-BS [57], and TGSR [26]. The eight baselines cover deep reconstruction, attention-based selection, clustering, and graph-based self-representation. Therefore, the comparison can provide a relatively comprehensive evaluation of the proposed method.

The number of superpixels is an important parameter that determines the granularity of superpixel segmentation. Following the adaptive strategy in [58], we determine the number of superpixels according to the spatial complexity of each scene as follows:

$$N_{\mathrm{sp}} = \left\lfloor T_{\mathrm{sp}} \frac{N_e}{N_{\mathrm{pix}}} + \frac{1}{2} \right\rfloor \tag{29}$$

where $\mathbf{E}$ is the binary edge map of the first principal component, $N_e = \|\mathbf{E}\|_0$ is the number of edge pixels, N_{pix} is the total number of pixels, and T_{sp} controls the superpixel granularity. A larger T_{sp} produces a finer partition.

All hyperparameters were selected from predefined discrete candidate sets. In particular, $\lambda \in \{1, 5, 10, 50, 100, 500, 1000\}$, whereas α, η, and $\gamma \in \{10^{-4}, 10^{-3}, \ldots, 10^{2}\}$. The structural settings T_{sp}, K_{nn}, σ, V, and k were varied over $\{500, 1000, \ldots, 6000\}$, $\{3, 4, \ldots, 15\}$, $\{0.1, 0.3, \ldots, 1.5\}$, $\{1, 2, \ldots, 10\}$, and $\{5, 10, \ldots, 60\}$, respectively. Sensitivity to the tradeoff and structural parameters is analyzed in Sections IV-D2 and IV-D1, respectively. Based on the structural sensitivity results, T_{sp}, K_{nn}, σ, V, and k were fixed to 3000, 5, 1.1, 4, and 20, respectively.

For quantitative evaluation, SVM, MLP, and RF are adopted to assess the classification performance of the selected band subsets. The SVM employed an RBF kernel, with C and γ determined by cross-validation. The MLP consisted of one hidden layer with 64 neurons and used ReLU activation and the Adam optimizer; the maximum number of iterations and the L_2 regularization coefficient were set to 500 and 10^{-4}, respectively. The RF comprised 50 decision trees.

On each dataset, 5% of the labeled samples from each class were randomly selected for training in the main experiments, while the remaining samples were used for testing. Additional experiments with training ratios of 2%, 3%, 4%, 5%, and 6%} were conducted to evaluate the classification performance of the selected band subsets under different limited-label conditions.To reduce the influence of random sampling, each experiment was repeated ten times independently, and the average results were reported. Overall accuracy (OA), average accuracy (AA), and Kappa coefficient are adopted as evaluation metrics. All compared methods are evaluated under the same training-testing protocol and classifier settings to ensure a fair comparison. All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU (24 GB memory) and a 12th Gen Intel Core i5-12600KF CPU.

B. Comparison With State-of-the-Art Methods

In this subsection, the proposed method is compared with eight representative unsupervised hyperspectral BS methods. The comparison is conducted from three aspects, including quantitative classification performance, classification map visualization, and selected band distribution analysis. For a fair comparison, all methods are evaluated under the same training-testing protocol and classifier settings. In particular, 10, 15, and 20 bands are selected for the IPs, PU, and SAs datasets, respectively, when reporting the detailed quantitative results.

1) Quantitative Classification Results: Fig. 2 shows the OA curves of different BS methods with different numbers of selected bands on the IPs, PU, and SAs datasets. Three commonly used classifiers, including MLP, SVM, and RF, are adopted for evaluation. Overall, the proposed method achieves competitive and stable classification performance across different datasets and classifiers. In particular, the proposed method maintains relatively high OA values when only a small number of bands are selected, indicating that it can effectively select informative bands while reducing spectral redundancy. In addition, the relatively stable OA curves indicate that the proposed method is not overly sensitive to the number of selected bands.

To further provide a detailed quantitative comparison, Tables II–IV report the per-class accuracy, OA, AA, and Kappa coefficient obtained by different methods using the SVM classifier. In particular, 10, 15, and 20 bands are selected for the IPs, PU, and SAs datasets, respectively. The best results are highlighted in bold.

As shown in Table II, the proposed method achieves the best OA, AA, and Kappa values of 72.83%, 67.53%, and 69.04%, respectively, on the IPs dataset. Compared with HLFC, the strongest competing method on this dataset, the proposed method improves OA, AA, and Kappa by 1.40, 1.07, and 1.66 percentage points, respectively. Since IPs contain many crop classes with similar spectral characteristics, these improvements indicate that the proposed method can better preserve discriminative spectral information and suppress redundant bands.

Table III reports the results on the PU dataset. The proposed method obtains the best OA, AA, and Kappa values of 91.59%, 89.21%, and 88.80%, respectively. Compared with DVS3C, which achieves the second-best overall performance, the proposed method improves OA, AA, and Kappa by 0.34, 0.86, and 0.46 percentage points, respectively. Although several methods already achieve high accuracies on this urban dataset, the proposed method still provides consistent gains, indicating the effectiveness of the selected bands for urban scene classification.

On the SAs dataset, as reported in Table IV, the proposed method achieves the best OA, AA, and Kappa values of 92.56%, 95.76%, and 91.71%, respectively. Since SAs have relatively regular spatial distributions and high-quality spectral information, the performance gap among different methods is limited. Nevertheless, the proposed method still maintains the best overall results with only 20 selected bands, demonstrating its robustness across different HSI scenes.

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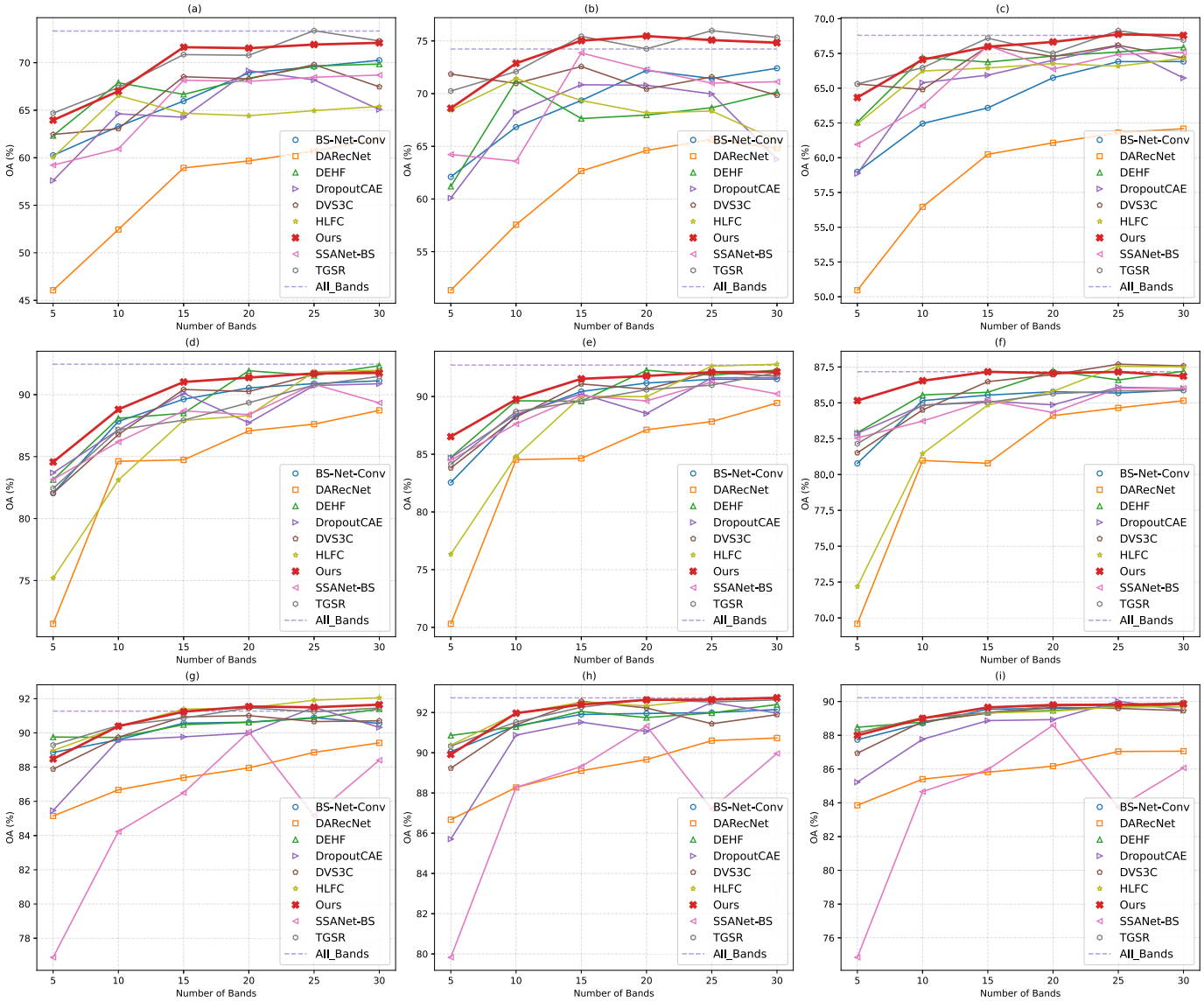


Fig. 2. OA comparison of different BS methods with different numbers of selected bands on the IPs, PU, and SAs datasets using MLP, SVM, and RF classifiers. (a) IP-MLP. (b) IP-SVM. (c) IP-RF. (d) PU-MLP. (e) PU-SVM. (f) PU-RF. (g) SA-MLP. (h) SA-SVM. (i) SA-RF.

In summary, both the OA curves and the detailed classification results demonstrate the effectiveness of the proposed method. The proposed method shows clear advantages on the challenging IPs dataset and maintains consistently strong performance on the PU and SAs datasets, indicating its ability to select compact, discriminative, and complementary band subsets.

2) Classification Performance Under Different Training Ratios: To evaluate the selected bands under limited-label conditions, the training ratio was varied over $\{2\%, 3\%, 4\%, 5\%, 6\%\}$. For each ratio, samples were randomly selected from each class, with at least one training sample retained per class, while the remaining samples were used for testing. The numbers of selected bands were fixed to 10, 15, and 20 for IP, PU, and SA, respectively. Each experiment was repeated ten times, and the mean OA obtained by the SVM classifier was reported.

As shown in Fig. 3, the OA of most methods increased as the training ratio increased. The proposed method achieved the highest or near-highest OA across the evaluated ratios on all three datasets. In particular, it remained competitive when only 2% of the labeled samples were available. These results demonstrated that the discriminative ability of the selected bands remained stable under limited-label conditions.

3) Visual Comparison of Classification Maps: To intuitively evaluate the quality of the selected bands, Figs. 4–6 show the classification maps obtained by different BS methods using the SVM classifier. The number of selected bands is fixed to 10, 15, and 20 for the IPs, PU, and SAs datasets, respectively.

As shown in Fig. 4, the IPs dataset contains complex agricultural regions with severe spectral similarity among different crop classes. Several compared methods generate noticeable salt-and-pepper noise and fragmented classification

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TABLE II

PER-CLASS ACCURACY, OA, AA, AND KAPPA OBTAINED BY DIFFERENT BS METHODS ON THE IPS DATASET USING SVM. THE NUMBER OF SELECTED BANDS IS FIXED TO 10. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Class	BS-Net-Conv	DARecNet	DEHF	DropoutCAE	DVS3C	HLFC	SSANet-BS	TGSR	Ours
1	54.55 ± 8.50	38.17 ± 15.11	34.09 ± 21.38	32.27 ± 19.31	**62.27 ± 12.51**	46.82 ± 23.92	8.64 ± 4.93	40.91 ± 18.53	46.36 ± 18.92
2	57.66 ± 3.08	48.87 ± 5.21	**68.99 ± 4.31**	62.79 ± 1.94	66.91 ± 4.05	61.72 ± 2.95	52.94 ± 4.37	68.81 ± 3.12	68.11 ± 1.78
3	58.65 ± 4.08	44.62 ± 4.46	60.33 ± 6.16	57.77 ± 4.55	57.28 ± 2.89	60.36 ± 2.92	42.54 ± 6.34	59.82 ± 5.77	**66.12 ± 2.46**
4	**44.80 ± 9.79**	13.78 ± 5.31	40.09 ± 12.03	44.18 ± 12.89	43.47 ± 10.06	36.89 ± 11.66	24.80 ± 6.75	44.00 ± 14.81	44.71 ± 13.99
5	82.92 ± 4.85	63.79 ± 2.54	82.40 ± 4.53	74.90 ± 5.62	78.69 ± 3.76	77.82 ± 4.19	71.85 ± 3.17	**84.92 ± 4.06**	80.65 ± 2.04
6	85.04 ± 1.90	81.44 ± 6.49	83.37 ± 3.35	87.06 ± 5.00	85.88 ± 5.06	90.92 ± 2.16	82.82 ± 3.67	84.32 ± 7.33	**92.02 ± 1.72**
7	34.81 ± 15.18	1.48 ± 3.31	59.26 ± 22.98	65.19 ± 21.37	57.04 ± 25.74	80.74 ± 10.28	19.26 ± 15.14	60.74 ± 14.49	**82.96 ± 9.30**
8	90.13 ± 2.41	91.85 ± 1.97	90.13 ± 4.78	92.16 ± 1.05	93.08 ± 2.74	**94.36 ± 3.50**	85.24 ± 6.08	93.30 ± 2.59	92.91 ± 2.65
9	18.95 ± 13.21	10.53 ± 7.44	25.26 ± 30.55	**27.37 ± 13.62**	25.26 ± 10.12	20.00 ± 5.77	9.47 ± 6.86	22.11 ± 13.62	17.89 ± 12.12
10	45.40 ± 4.82	37.77 ± 3.57	64.59 ± 6.18	60.85 ± 4.02	65.61 ± 3.38	69.69 ± 4.18	51.55 ± 4.67	67.69 ± 3.74	**70.49 ± 1.72**
11	70.33 ± 3.54	59.80 ± 3.37	69.21 ± 3.34	69.02 ± 3.14	72.18 ± 1.08	**73.75 ± 2.14**	58.97 ± 1.39	70.93 ± 1.44	73.21 ± 1.28
12	50.37 ± 2.81	32.22 ± 4.01	46.15 ± 5.53	49.77 ± 2.69	38.86 ± 5.39	**55.17 ± 6.31**	33.43 ± 4.91	51.79 ± 10.03	53.46 ± 4.93
13	92.51 ± 1.29	90.56 ± 6.49	**93.03 ± 3.11**	90.36 ± 7.97	83.49 ± 9.79	86.67 ± 6.04	84.51 ± 10.67	90.67 ± 3.67	85.13 ± 8.93
14	89.52 ± 3.55	85.86 ± 2.56	**90.20 ± 3.44**	84.34 ± 3.39	88.89 ± 6.38	87.67 ± 3.68	83.96 ± 2.85	87.67 ± 4.42	87.39 ± 2.30
15	31.72 ± 7.77	26.54 ± 5.08	32.15 ± 6.10	**35.37 ± 7.47**	33.13 ± 4.64	33.46 ± 7.17	22.40 ± 4.70	33.19 ± 10.75	31.77 ± 7.16
16	70.00 ± 9.93	61.82 ± 13.12	87.05 ± 3.82	82.95 ± 4.02	82.50 ± 3.07	87.27 ± 2.94	80.45 ± 6.19	**87.95 ± 2.06**	87.27 ± 3.89
OA	67.16 ± 1.13	57.77 ± 0.73	70.18 ± 1.56	68.34 ± 1.55	70.02 ± 1.35	71.43 ± 1.86	59.34 ± 1.23	71.29 ± 1.67	**72.83 ± 1.09**
AA	61.09 ± 1.92	49.32 ± 1.36	64.14 ± 3.89	63.52 ± 2.82	64.66 ± 1.33	66.46 ± 2.24	50.80 ± 1.82	65.55 ± 3.17	**67.53 ± 2.49**
Kappa	62.36 ± 1.30	51.43 ± 0.85	66.01 ± 1.72	63.87 ± 1.73	65.73 ± 1.49	67.38 ± 2.14	53.72 ± 1.49	67.28 ± 1.92	**69.04 ± 1.27**

TABLE III

PER-CLASS ACCURACY, OA, AA, AND KAPPA OBTAINED BY DIFFERENT BS METHODS ON THE PU DATASET USING SVM. THE NUMBER OF SELECTED BANDS IS FIXED TO 15. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Class	BS-Net-Conv	DARecNet	DEHF	DropoutCAE	DVS3C	HLFC	SSANet-BS	TGSR	Ours
1	89.92 ± 1.55	88.08 ± 0.59	91.01 ± 1.13	91.17 ± 0.91	90.76 ± 1.20	89.07 ± 1.23	90.49 ± 0.63	**91.71 ± 0.99**	91.02 ± 1.45
2	96.64 ± 0.23	94.71 ± 0.65	95.84 ± 0.39	96.40 ± 0.36	96.74 ± 0.44	**97.27 ± 0.31**	96.72 ± 0.25	96.49 ± 0.19	96.77 ± 0.36
3	70.92 ± 2.68	57.38 ± 3.13	71.74 ± 1.92	72.57 ± 1.66	73.80 ± 1.91	60.85 ± 1.33	68.10 ± 2.26	69.52 ± 2.54	**74.89 ± 2.09**
4	88.11 ± 1.18	75.89 ± 2.53	90.57 ± 1.33	88.11 ± 1.00	90.20 ± 1.69	90.88 ± 0.77	91.10 ± 1.25	**92.52 ± 1.11**	90.87 ± 0.72
5	99.11 ± 0.12	98.98 ± 0.20	99.12 ± 0.16	99.14 ± 0.16	98.97 ± 0.22	99.23 ± 0.27	**99.28 ± 0.18**	99.14 ± 0.12	98.94 ± 0.31
6	78.99 ± 1.34	63.93 ± 1.85	72.86 ± 1.72	78.46 ± 1.16	**84.08 ± 1.56**	82.64 ± 1.02	74.86 ± 1.76	70.43 ± 1.60	83.41 ± 0.83
7	75.97 ± 2.42	72.15 ± 1.33	79.72 ± 2.72	**82.67 ± 2.66**	74.92 ± 1.90	68.45 ± 4.52	77.18 ± 2.45	79.38 ± 4.38	80.02 ± 3.30
8	87.02 ± 1.82	77.83 ± 3.86	84.80 ± 1.55	**87.34 ± 1.38**	85.76 ± 1.69	82.45 ± 0.88	86.23 ± 1.04	85.67 ± 1.64	87.04 ± 2.14
9	99.80 ± 0.18	99.22 ± 0.62	99.87 ± 0.18	99.87 ± 0.18	**99.93 ± 0.10**	99.89 ± 0.00	99.91 ± 0.20	99.89 ± 0.14	99.91 ± 0.09
OA	90.33 ± 0.22	84.97 ± 0.15	89.57 ± 0.26	90.67 ± 0.20	91.25 ± 0.38	89.98 ± 0.22	90.02 ± 0.28	89.78 ± 0.18	**91.59 ± 0.22**
AA	87.38 ± 0.19	80.91 ± 0.35	87.28 ± 0.44	88.41 ± 0.36	88.35 ± 0.53	85.64 ± 0.66	87.10 ± 0.30	87.20 ± 0.41	**89.21 ± 0.53**
Kappa	87.08 ± 0.29	79.79 ± 0.24	86.06 ± 0.36	87.55 ± 0.27	88.34 ± 0.50	86.63 ± 0.30	86.65 ± 0.38	86.31 ± 0.25	**88.80 ± 0.29**

TABLE IV

PER-CLASS ACCURACY, OA, AA, AND KAPPA OBTAINED BY DIFFERENT BS METHODS ON THE SAS DATASET USING SVM. THE NUMBER OF SELECTED BANDS IS FIXED TO 20. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Class	BS-Net-Conv	DARecNet	DEHF	DropoutCAE	DVS3C	HLFC	SSANet-BS	TGSR	Ours
1	99.21 ± 0.47	97.88 ± 0.42	98.65 ± 1.25	98.46 ± 0.90	99.43 ± 0.43	99.35 ± 0.38	99.25 ± 0.56	99.33 ± 0.53	**99.50 ± 0.40**
2	99.79 ± 0.10	98.56 ± 0.55	99.68 ± 0.08	99.69 ± 0.18	99.82 ± 0.06	99.80 ± 0.16	99.73 ± 0.11	**99.85 ± 0.09**	99.80 ± 0.13
3	97.72 ± 0.95	97.35 ± 1.14	**98.65 ± 0.41**	97.10 ± 1.47	98.34 ± 0.53	97.69 ± 1.73	97.24 ± 1.13	97.95 ± 1.28	98.53 ± 0.52
4	98.46 ± 1.27	99.17 ± 0.43	98.82 ± 0.76	99.02 ± 0.45	99.27 ± 0.53	99.27 ± 0.30	**99.29 ± 0.38**	99.24 ± 0.58	99.18 ± 0.42
5	97.39 ± 1.38	95.65 ± 1.53	**98.33 ± 0.69**	97.76 ± 1.07	98.15 ± 0.71	97.45 ± 1.51	97.90 ± 1.04	97.96 ± 0.88	98.00 ± 1.10
6	**99.81 ± 0.06**	99.30 ± 0.25	99.79 ± 0.04	**99.81 ± 0.05**	99.80 ± 0.06	99.79 ± 0.09	99.76 ± 0.08	99.79 ± 0.04	99.77 ± 0.09
7	99.71 ± 0.14	99.19 ± 0.34	99.64 ± 0.19	99.62 ± 0.14	99.71 ± 0.16	99.75 ± 0.15	99.60 ± 0.11	99.68 ± 0.19	**99.76 ± 0.11**
8	87.27 ± 1.00	86.03 ± 1.36	87.07 ± 1.55	86.74 ± 1.27	87.75 ± 1.37	88.53 ± 1.26	86.00 ± 1.32	87.98 ± 1.10	**88.59 ± 0.77**
9	**99.70 ± 0.39**	97.90 ± 0.28	99.66 ± 0.24	99.42 ± 0.43	99.56 ± 0.24	99.65 ± 0.24	99.62 ± 0.16	99.62 ± 0.58	99.65 ± 0.33
10	92.25 ± 0.63	88.79 ± 1.08	92.88 ± 1.14	90.92 ± 1.78	93.93 ± 0.73	91.25 ± 0.61	92.77 ± 1.19	**94.27 ± 1.24**	93.50 ± 0.54
11	93.77 ± 2.19	79.78 ± 0.90	93.77 ± 2.18	89.44 ± 4.09	95.19 ± 2.14	90.58 ± 2.38	93.85 ± 2.71	**95.45 ± 2.42**	93.46 ± 4.26
12	98.70 ± 2.27	95.61 ± 0.92	99.26 ± 0.85	98.85 ± 0.58	99.72 ± 0.26	99.62 ± 0.47	99.42 ± 1.06	99.65 ± 0.29	**99.74 ± 0.27**
13	97.93 ± 1.50	97.13 ± 1.71	97.72 ± 0.95	96.57 ± 2.10	99.08 ± 1.19	**99.13 ± 1.09**	98.60 ± 0.89	98.21 ± 1.18	98.99 ± 1.20
14	95.24 ± 1.65	92.62 ± 1.80	93.41 ± 2.35	90.77 ± 3.38	**96.30 ± 0.67**	95.91 ± 1.60	95.85 ± 0.78	94.13 ± 3.45	95.75 ± 1.96
15	67.37 ± 1.17	61.01 ± 1.38	66.46 ± 0.93	64.68 ± 1.85	65.59 ± 0.94	67.88 ± 1.19	64.69 ± 2.21	67.45 ± 1.63	**69.45 ± 0.86**
16	97.98 ± 0.17	97.18 ± 0.57	97.89 ± 0.33	97.40 ± 0.81	**98.87 ± 0.09**	98.68 ± 0.22	98.49 ± 0.31	98.54 ± 0.16	98.56 ± 0.44
OA	91.77 ± 0.14	89.48 ± 0.13	91.67 ± 0.22	90.95 ± 0.12	91.94 ± 0.25	92.09 ± 0.23	91.25 ± 0.17	92.17 ± 0.15	**92.56 ± 0.12**
AA	95.14 ± 0.27	92.70 ± 0.14	95.11 ± 0.28	94.14 ± 0.39	95.66 ± 0.22	95.27 ± 0.22	95.13 ± 0.24	95.57 ± 0.41	**95.76 ± 0.39**
Kappa	90.82 ± 0.16	88.25 ± 0.14	90.71 ± 0.25	89.90 ± 0.13	91.01 ± 0.28	91.18 ± 0.26	90.24 ± 0.19	91.27 ± 0.17	**91.71 ± 0.13**

regions. For example, DARecNet and SSANet-BS produce many scattered misclassified pixels, while BS-Net-Conv and DropoutCAE also suffer from local confusion in several crop areas. On the contrary, the proposed method obtains a cleaner classification map with high regional consistency and achieves the highest OA of 74.42%. This result indicates that the

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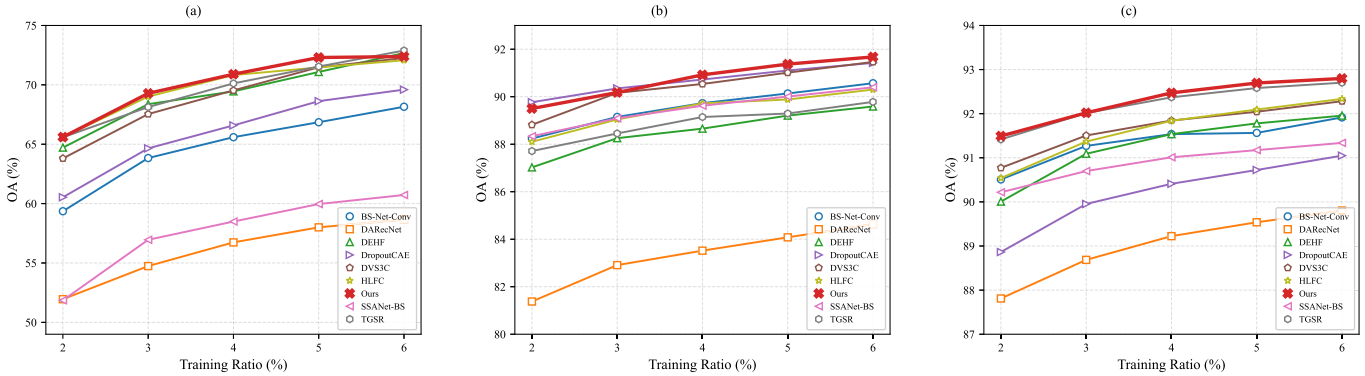


Fig. 3. OA comparison under different training ratios on (a) IPs, (b) PU, and (c) SAs. The numbers of selected bands were fixed to 10, 15, and 20, respectively.

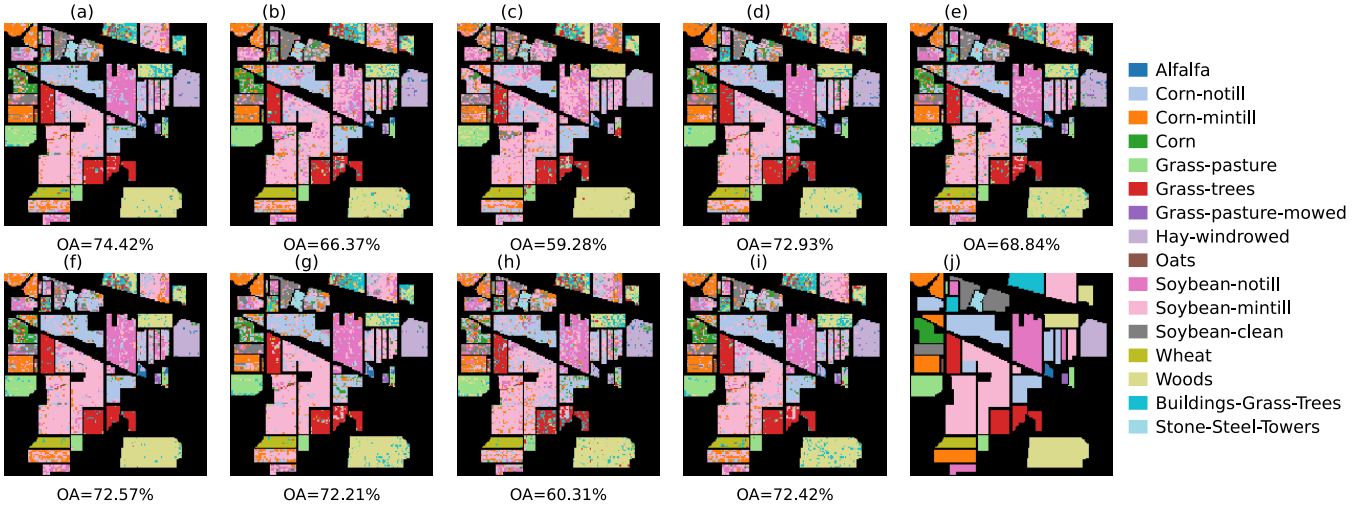


Fig. 4. Classification maps of different BS methods on the IPs dataset using SVM with ten selected bands. (a) Ours. (b) BS-Net-Conv. (c) DARecNet. (d) DEHF. (e) DropoutCAE. (f) DVS3C. (g) HLFC. (h) SSANet-BS. (i) TGSR. (j) Ground Truth.

selected bands are effective for distinguishing spectrally similar land-cover classes.

Fig. 5 presents the visual results on the PU dataset. Since this dataset contains urban objects with clear spatial structures, most methods can preserve the main land-cover regions. However, some comparison methods still show local misclassification around roads, bare soil, and shadow areas. The proposed method produces a more accurate classification map and achieves the highest OA of 91.34%. The boundaries of major regions are better preserved, and the visual result is closer to the ground truth, demonstrating the effectiveness of the selected bands for urban scene classification.

For the SAs dataset, as shown in Fig. 6, all methods obtain relatively smooth classification maps due to the regular spatial distribution of agricultural fields. Nevertheless, the proposed method still achieves the best OA of 92.66%, slightly outperforming TGSR and other competitors. Compared with several methods that contain scattered errors inside large homogeneous regions, the proposed method better maintains the integrity of crop areas and provides clearer class boundaries. These visual results further confirm that the proposed method can select representative bands for different HSI scenes.

4) Analysis of Selected Bands: Figs. 7–9 show the selected band positions of different methods and the entropy value of each band on the three datasets. Entropy is used as an auxiliary indicator to reflect the information content of each band.

As shown in the figures, several competing methods select bands clustered within narrow spectral regions, which may lead to information redundancy because adjacent bands are highly correlated. On the contrary, the proposed method selects bands from broader spectral ranges on all three datasets, indicating that the selected subset contains more complementary spectral information.

Meanwhile, most selected bands of the proposed method are located near informative spectral regions with relatively high entropy values. It is worth noting that they are not always exactly located at the peaks of the entropy curves. This is reasonable because entropy only measures the information content of individual bands, while effective BS should also consider interband redundancy and structural complementarity. Therefore, the proposed method achieves a better balance between informativeness and redundancy reduction, which contributes to its stable classification performance.

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Fig. 5. Classification maps of different BS methods on the PU dataset using SVM with 15 selected bands. (a) Ours. (b) BS-Net-Conv. (c) DEHF. (d) DARecNet. (e) DVS3C. (f) DropoutCAE. (g) HLFC. (h) SSANet-BS. (i) TGSR. (j) Ground Truth.

C. Ablation Study

To investigate the contributions of the key components, four variants of the proposed method are constructed. The first variant, denoted as *First-order graph*, replaces the high-order graph prior $f(\mathbf{C})$ with the original first-order graph $\mathbf{C}$. The second variant, denoted as *w/o orthogonal regularization*, removes the orthogonal regularization imposed on the projection matrix $\mathbf{H}$. The third variant, denoted as *w/o Laplacian regularization*, removes the Laplacian regularization term. The fourth variant, denoted as *w/o graph consistency*, removes the graph-consistency term by setting $\gamma = 0$ while retaining the high-order Laplacian regularization with $\eta > 0$ and $V = 4$. The complete method is denoted as Full model.

The self-representation term and the $\ell_{2,1}$-norm regularization are retained in all variants because they constitute the fundamental BS mechanism of the proposed method. For consistency with the quantitative comparison, 10, 15, and 20 bands are selected for the IPs, PU, and SAs datasets, respectively, and OA is reported as the evaluation metric.

As shown in Table V, the full model achieves the highest OA on all three datasets. Compared with the *First-order graph* variant, it improves OA by 2.04, 4.73, and 0.93 percentage points on IP, PU, and SA, respectively, confirming the benefit of modeling multistep neighborhood relationships. Removing the orthogonal regularization results in reductions of 16.84, 6.74, and 5.14 percentage points, indicating its importance in reducing correlation among projection directions and stabilizing sparse projection learning. Removing the Laplacian regularization decreases OA by 5.67, 5.97, and 3.35 percentage points, demonstrating its contribution to preserving local neighborhood relationships. Removing the graph-consistency term decreases OA by 5.52, 5.98, and 3.46 percentage points, respectively. These reductions demonstrate that guiding the learned self-representation matrix $\mathbf{S}$ toward the high-order graph prior $\mathbf{F}$ contributes substantially to the overall performance. Overall, the results verify that high-order graph propagation, graph consistency, orthogonal regularization, and Laplacian regularization each contribute to the performance of the proposed method.

TABLE V
ABLATION RESULTS ON THREE HSI DATASETS. ONLY OA VALUES (%) ARE REPORTED

Method	IP (10)	PU (15)	SA (20)
First-order graph	70.79	86.86	91.63
w/o Orthogonal Reg.	55.99	84.85	87.42
w/o Laplacian Reg.	67.16	85.62	89.21
w/o Graph Consistency	67.31	85.61	89.10
Full model	**72.83**	**91.59**	**92.56**

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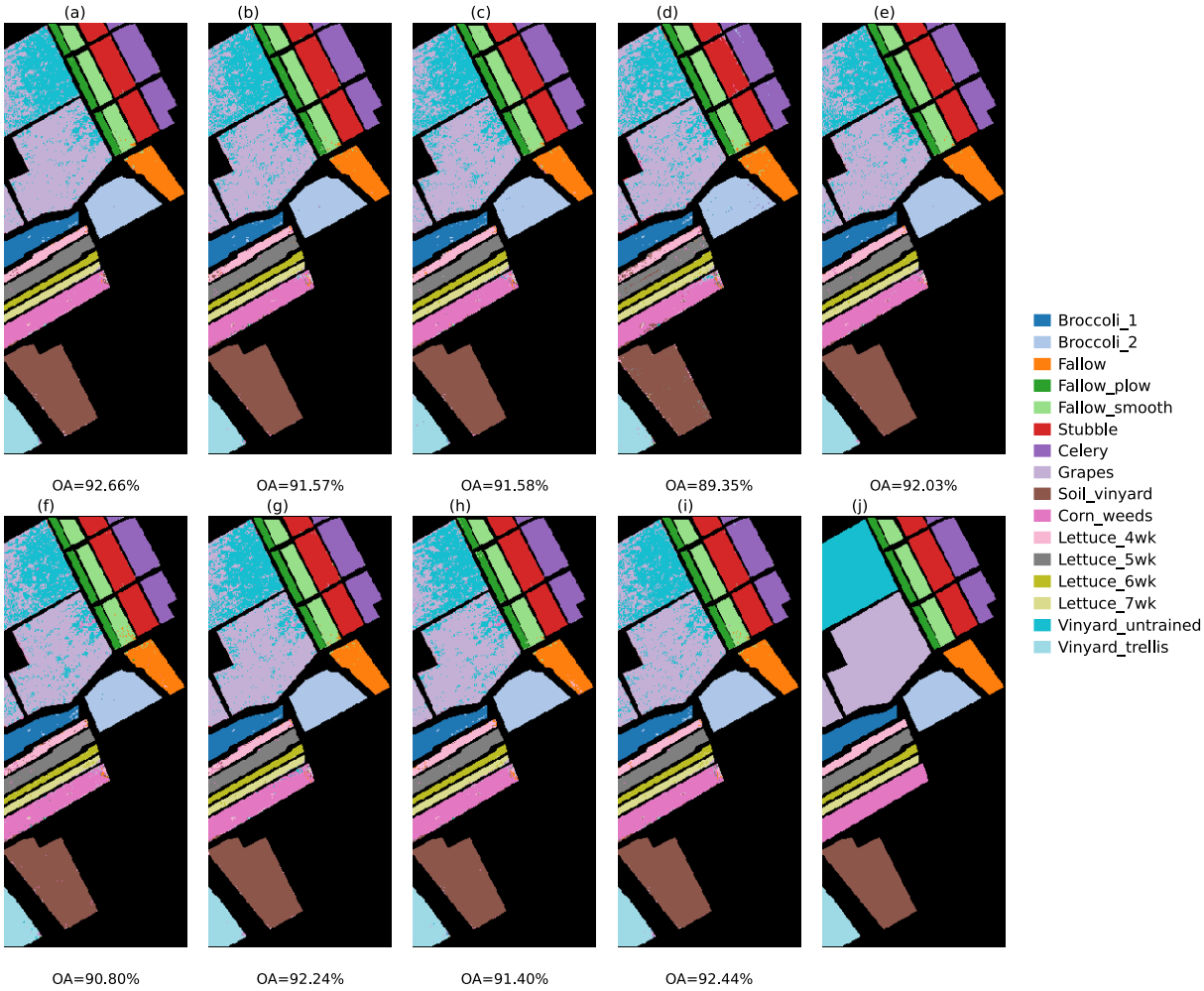


Fig. 6. Classification maps of different BS methods on the SAs dataset using SVM with 20 selected bands. (a) Ours. (b) BS-Net-Conv. (c) DEHF. (d) DARecNet. (e) DVS3C. (f) DropoutCAE. (g) HLFC. (h) SSANet-BS. (i) TGSR. (j) Ground Truth.

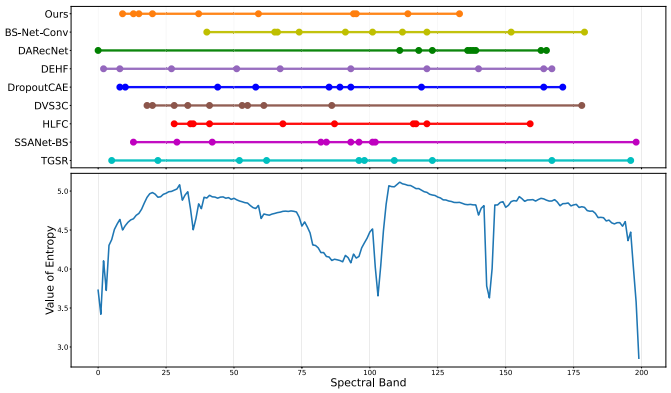


Fig. 7. Best ten bands of the IPs dataset selected by (Top) different BS methods and (Bottom) entropy value of each band.

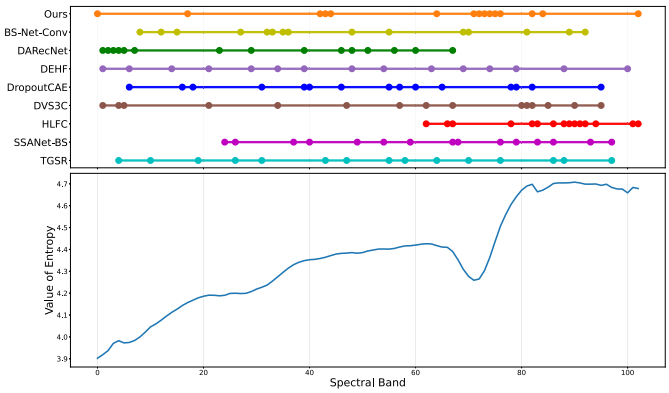


Fig. 8. Best 15 bands of the PU dataset selected by (Top) different BS methods and (Bottom) entropy value of each band.

D. Model Analysis

1) Sensitivity to Structural Settings: To examine the sensitivity of the proposed method to structural settings, we investigate the superpixel granularity coefficient T_{sp}, the KNN neighborhood size K_{nn}, the kernel bandwidth σ, the propagation order V, and the latent dimension k. An 20 bands are selected for each dataset, and the OA obtained by the SVM classifier is used as the evaluation metric. Following a one-factor-at-a-time protocol, each setting is varied while the remaining settings are fixed. The same evaluation protocol as

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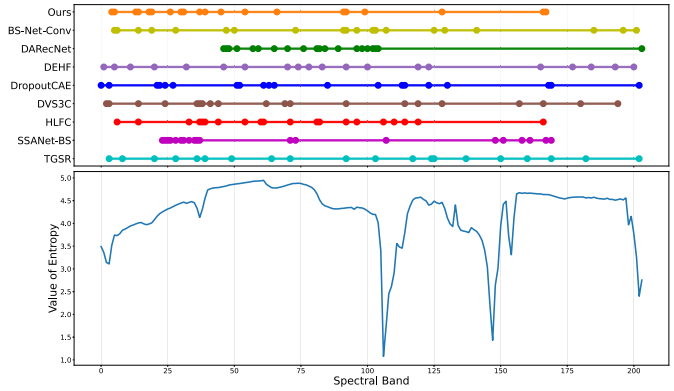


Fig. 9. Best 20 bands of the SAs dataset selected by (top) different BS methods and (bottom) entropy value of each band.

TABLE VI

END-TO-END RUNNING TIMES (S) OF DIFFERENT METHODS ON THREE HSI DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Indian Pines	Pavia University	Salinas
BS-Net-Conv	68.632	498.649	364.620
DARecNet	488.450	1167.570	2811.470
DEHF	16.586	187.133	186.226
DropoutCAE	3.472	14.738	19.457
DVS3C	8.535	19.540	25.105
HLFC	**1.151**	**1.567**	**1.626**
SSANet-BS	58.755	341.943	345.847
TGSR	8.484	17.948	51.675
Ours	<u>2.684</u>	<u>3.833</u>	<u>5.596</u>

that used in the main experiments is adopted for all parameter configurations.

Fig. 10(a) shows the influence of the superpixel granularity coefficient T_{sp}. The OA on IP generally increases with T_{sp}, although moderate fluctuations are observed. PU and SA maintain high OA values over a broad range and perform favorably at intermediate granularities. A small T_{sp} results in fewer superpixels and may mix heterogeneous pixels within the same region, thereby reducing regional homogeneity. Conversely, an excessively large T_{sp} produces a finer partition and more region-level samples, thereby increasing the costs of graph construction and alternating optimization. It may also weaken the noise-suppression benefit provided by regional averaging. Considering both classification performance and computational efficiency, T_{sp} is set to 3000.

As shown in Fig. 10(b), the three datasets achieve relatively high OA values when K_{nn} is small. Increasing the neighborhood size generally reduces performance on PU and SA and causes more noticeable fluctuations on IP. A large neighborhood may connect spectrally dissimilar regions and weaken the discriminative local structure. Therefore, K_{nn} is set to 5 to balance graph connectivity and local structure preservation.

Fig. 10(c) shows that the OA values remain stable when σ varies from 0.3 to 1.5. Only $\sigma = 0.1$ yields relatively lower performance on IP and PU. This result indicates that the proposed method is insensitive to the kernel bandwidth within a broad range, and σ is set to 1.1.

The effect of the propagation order V is presented in Fig. 10(d). On IP, the OA increases from $V = 1$ to $V = 4$, demonstrating the benefit of incorporating moderate multistep structural information. SA remains stable across different propagation orders, while PU maintains high performance for $V \leq 5$ but declines at larger orders. These results indicate that moderate propagation enriches structural dependencies without substantially weakening local neighborhoods. Excessive propagation may diffuse the graph structure and lead to oversmoothing. Therefore, V is set to 4.

Fig. 10(e) presents the influence of the latent dimension k. IP achieves higher OA values when k ranges from 15 to 25 and reaches its best performance near $k = 20$. PU and SA remain stable over a broader range, although their performance decreases when k becomes excessively large. A small latent dimension may limit representation capacity, whereas an overly large dimension may introduce redundant projection directions and increase computational cost. Accordingly, k is set to 20.

Overall, the proposed method remains stable within reasonable ranges of the five structural settings, although extreme configurations can reduce performance on individual datasets. The adopted settings are selected from consistently high-performing regions across the three datasets rather than isolated dataset-specific optima.

2) Sensitivity to Tradeoff Parameters: To evaluate the sensitivity of the proposed method to parameter variations, four tradeoff parameters, i.e., α, λ, η, and γ, are investigated on the IP, PU, and SA datasets. Since directly varying all four parameters would lead to a high-dimensional parameter space that is difficult to visualize, we divide them into two functionally related groups according to their roles in the objective function. In particular, α and λ are associated with the row-sparsity constraint and the soft orthogonal regularization term, respectively, and mainly affect the learning of the sparse projection matrix. On the contrary, η and γ control the high-order Laplacian structure-preserving term and the high-order graph consistency term, respectively, and mainly influence structural information preservation.

To keep consistency with the main experiments, the number of selected bands is fixed to 10, 15, and 20 for the IP, PU, and SA datasets, respectively. The OA obtained by the SVM classifier is used as the evaluation metric. As shown in Fig. 11(a)–(c) shows the OA values under different combinations of α and λ with η and γ fixed, while (d)–(f) shows the OA values under different combinations of η and γ with α and λ fixed.

It can be observed that the OA values on the PU and SA datasets remain relatively stable under different parameter combinations. In comparison, the IP dataset shows more noticeable fluctuations, which may be related to its limited training samples and complex class distribution. When some parameters are set to excessively large or small values, the classification performance decreases, indicating that a proper balance among different regularization terms is necessary. Overall, the proposed method maintains stable performance within a reasonable parameter range, demonstrating its robustness to parameter settings.

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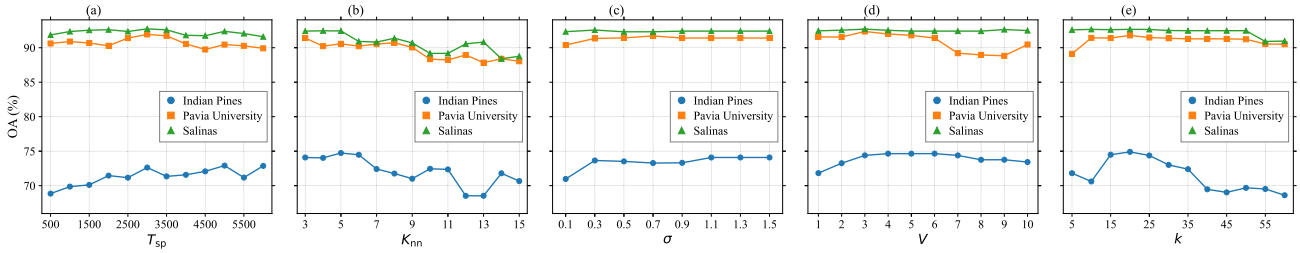


Fig. 10. Sensitivity of the proposed method to structural settings on the IP, PU, and SA datasets. The investigated settings include (a) superpixel granularity coefficient T_{sp}, (b) KNN neighborhood size K_{nn}, (c) kernel bandwidth σ, (d) propagation order V, and (e) latent dimension k. An 20 bands are selected on each dataset, and one setting is varied at a time while the remaining settings are fixed. Each point reports the mean OA obtained by the SVM classifier over ten independent runs.

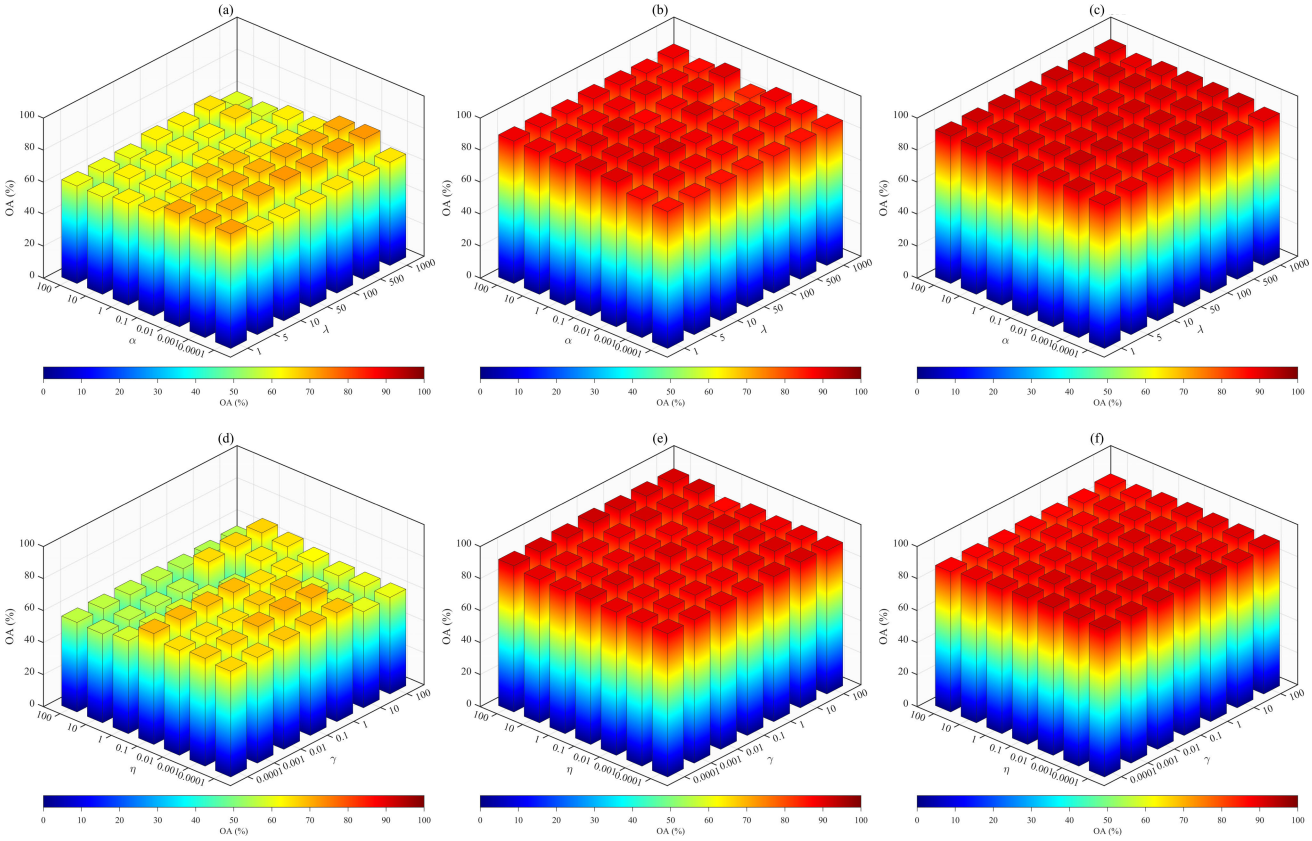
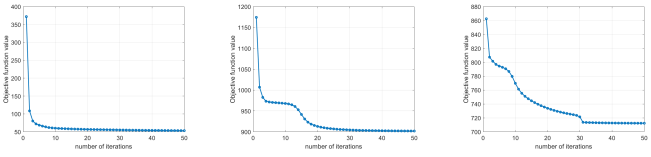


Fig. 11. Parameter sensitivity analysis of the proposed method on the IP, PU, and SA datasets. (a)–(c) OA values obtained by varying α and λ with η and γ fixed. (d)–(f) OA values obtained by varying η and γ with α and λ fixed. The number of selected bands is fixed to 10, 15, and 20 for IP, PU, and SA, respectively.

3) Running Time Analysis: Table VI reports the end-to-end running times of different methods under the hardware settings described above. HLFC achieves the shortest running time, while the proposed method ranks second on all three datasets. In particular, the proposed method requires 2.684, 3.833, and 5.596 s on IP, PU, and SA, respectively. Compared with all baselines except HLFC, the proposed method reduces running time by 22.69%–99.80% across the 21 method-dataset comparisons. This efficiency mainly results from operating on region-level samples without requiring network training. Since the methods retain their original CPU- or GPU-based implementations, these results represent practical end-to-end running times rather than hardware-normalized comparisons.

4) Convergence Analysis: Fig. 12 shows the convergence curves of the proposed method on the IP, PU, and SA datasets.



(a) (b) (c)

Fig. 12. Convergence curves of the proposed method on three different datasets. (a) IPs. (b) PU. (c) SAs.

The objective function value decreases rapidly in the early iterations and then gradually becomes stable, which demonstrates the convergence behavior of the proposed optimization algorithm.

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V. Conclusion

This article proposed a high-order graph-regularized sparse projection method for unsupervised hyperspectral BS. It represents an HSI using region-level samples and constructs a high-order graph to encode multistep relationships among them. Graph consistency and Laplacian regularization then integrate this structural prior with self-representation and row-sparse projection, enabling explicit estimation of band importance.

Experiments on three public HSI datasets show that the proposed method selects compact and discriminative band subsets, achieving competitive classification performance against eight representative methods, particularly under small band budgets. Selected-band analysis, ablation experiments, sensitivity tests, and running-time comparisons further support the effectiveness, robustness, and efficiency of the framework.

Several limitations remain. First, the high-order graph is constructed from a predefined neighborhood structure and propagation order and remains fixed during optimization. Its performance may therefore be affected by inaccuracies in the initial neighborhood structure, while excessive propagation may lead to structural oversmoothing. Second, the region-level representation relies on superpixel segmentation and mean spectra, which may lose fine-grained spatial–spectral information, particularly around complex boundaries and mixed regions. Future work will explore adaptive high-order graph construction jointly with sparse projection and develop multiscale, boundary-aware representations that combine pixel- and region-level information.

References

[1] G. Shaw and D. Manolakis, "Signal processing for hyperspectral image exploitation," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 12–16, Jan. 2002.

[2] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[3] D. Wang, L. Zhuang, L. Gao, X. Sun, X. Zhao, and A. Plaza, "Sliding dual-window-inspired reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504115.

[4] D. Wang, L. Ren, X. Sun, L. Gao, and J. Chanussot, "Nonlocal and local feature-coupled self-supervised network for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6981–6993, 2025.

[5] H. Zhang, H. Sun, H. Gao, L. Gao, and B. Zhang, "Hyperspectral remote sensing object detection via cross-domain learning from visible images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5510518.

[6] L. M. Bruce, C. H. Koger, and J. Li, "Dimensionality reduction of hyperspectral data using discrete wavelet transform feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2331–2338, Jan. 2002.

[7] Y. Zhou, J. Peng, and C. L. P. Chen, "Dimension reduction using spatial and spectral regularized local discriminant embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1082–1095, Feb. 2015.

[8] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.

[9] X. Zhang, X. Jiang, J. Jiang, Y. Zhang, X. Liu, and Z. Cai, "Spectral–spatial and superpixelwise PCA for unsupervised feature extraction of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502210.

[10] Q. Wang, F. Zhang, and X. Li, "Hyperspectral band selection via optimal neighborhood reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8465–8476, Dec. 2020.

[11] Y. Zhang, X. Wang, X. Jiang, and Y. Zhou, "Marginalized graph self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516712.

[12] M. Gong, M. Zhang, and Y. Yuan, "Unsupervised band selection based on evolutionary multiobjective optimization for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 544–557, Jan. 2016.

[13] X. Ou, M. Wu, B. Tu, G. Zhang, and W. Li, "Multi-objective unsupervised band selection method for hyperspectral images classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1952–1965, 2023.

[14] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[15] L. Wang, H.-C. Li, B. Xue, and C.-I. Chang, "Constrained band subset selection for hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2032–2036, Nov. 2017.

[16] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.

[17] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[18] Q. Wang, Q. Li, and X. Li, "Hyperspectral band selection via adaptive subspace partition strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4940–4950, Dec. 2019.

[19] Q. Li, X. Luo, and Y. Wang, "Multigranularity manifold hybrid adaptive near-neighbor preserved clustering for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5517020.

[20] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Laplacian-regularized low-rank subspace clustering for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1723–1740, Mar. 2019.

[21] Y. Cai, Z. Zhang, X. Liu, and Z. Cai, "Efficient graph convolutional self-representation for band selection of hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4869–4880, 2020.

[22] Y. Zhang, X. Wang, X. Jiang, and Y. Zhou, "Robust dual graph self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538513.

[23] C. Tang, X. Liu, E. Zhu, L. Wang, and A. Zomaya, "Hyperspectral band selection via spatial–spectral weighted region-wise multiple graph fusion-based spectral clustering," in *Proc. Thirtieth Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 3038–3044.

[24] J. Wang, C. Tang, X. Zheng, X. Liu, W. Zhang, and E. Zhu, "Graph regularized spatial–spectral subspace clustering for hyperspectral band selection," *Neural Netw.*, vol. 153, pp. 292–302, Sep. 2022.

[25] C. Tang et al., "Spatial and spectral structure preserved self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531413.

[26] Y. Zhang, J. Qi, X. Wang, Z. Cai, J. Peng, and Y. Zhou, "Tensorial global-local graph self-representation for hyperspectral band selection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13213–13225, Dec. 2024.

[27] X. Tian, C. Wang, L. Guo, and Y. Li, "Band selection for HSI classification using binary constrained optimization," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] X. Geng, K. Sun, L. Ji, and Y. Zhao, "A fast volume-gradient-based band selection method for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7111–7119, Nov. 2014.

[29] L. Zhang, Y. Zhong, B. Huang, J. Gong, and P. Li, "Dimensionality reduction based on clonal selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4172–4186, Dec. 2007.

[30] F. Xie, F. Li, C. Lei, J. Yang, and Y. Zhang, "Unsupervised band selection based on artificial bee colony algorithm for hyperspectral image classification," *Appl. Soft Comput.*, vol. 75, pp. 428–440, Feb. 2019.

[31] M. Wang, W. Liu, M. Chen, X. Huang, and W. Han, "A band selection approach based on a modified gray wolf optimizer and weight updating of bands for hyperspectral image," *Appl. Soft Comput.*, vol. 112, Nov. 2021, Art. no. 107805.

[32] M. Zhang, M. Gong, and Y. Chan, "Hyperspectral band selection based on multi-objective optimization with high information and low redundancy," *Appl. Soft Comput.*, vol. 70, pp. 604–621, Sep. 2018.

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[33] M. Wu et al., "Heterogeneous cuckoo search-based unsupervised band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500616.

[34] Y. Wan, C. Chen, A. Ma, L. Zhang, X. Gong, and Y. Zhong, "Adaptive multistrategy particle swarm optimization for hyperspectral remote sensing image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520115.

[35] Q. Wang et al., "Unsupervised hyperspectral band selection via structure-conserved and neighborhood-grouped evolutionary algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522319.

[36] H. Jia and Z. Li, "Hybrid multistrategy remora optimization algorithm-based band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531716.

[37] B. Xu, X. Li, W. Hou, Y. Wang, and Y. Wei, "A similarity-based ranking method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9585–9599, Nov. 2021.

[38] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, *A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification*, *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[39] L. Li, J. Zhou, Q. Zhang, and M. Song, *An information entropy feature preservation-based band selection and noise reduction method for hyperspectral image classification*, *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5507517.

[40] H. Zhang, H. Sun, X. Sun, H. Gao, L. Gao, and B. Zhang, "Spectral–spatial out-of-distribution-based unsupervised band selection method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538915.

[41] A. Martínez-Usó, F. Pla, J. M. Sotoca, and P. García-Sevilla, "Clustering-based hyperspectral band selection using information measures," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4158–4171, Dec. 2007.

[42] J. Wang et al., "Hyperspectral band selection via region-aware latent features fusion based clustering," *Inf. Fusion*, vol. 79, pp. 162–173, Mar. 2022.

[43] J. Feng et al., "Convolutional neural network based on bandwise-independent convolution and hard thresholding for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4414–4428, Sep. 2021.

[44] L. Mou, S. Saha, Y. Hua, F. Bovolo, L. Bruzzone, and X. X. Zhu, "Deep reinforcement learning for band selection in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504414.

[45] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020.

[46] K. He et al., "A dual global–local attention network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527613.

[47] J. Feng et al., "Multi-agent deep reinforcement learning for hyperspectral band selection with hybrid teacher guide," *Knowl.-Based Syst.*, vol. 299, Sep. 2024, Art. no. 112044. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950705124006786

[48] J. Feng, G. Bai, D. Li, X. Zhang, R. Shang, and L. Jiao, "MR-selection: A meta-reinforcement learning approach for zero-shot hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500320.

[49] Y. Liu, X. Li, Z. Xu, and Z. Hua, "BSFormer: Transformer-based reconstruction network for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[50] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.

[51] F. Nie, H. Huang, X. Cai, and C. Ding, "Efficient and robust feature selection via joint l2,1-norms minimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 23, 2010, pp. 1813–1821. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2010/file/09c6c3783b4a70054da74f2538ed47c6-Paper.pdf

[52] S. K. Roy, S. Das, T. Song, and B. Chanda, "DARecNet-BS: Unsupervised dual-attention reconstruction network for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 12, pp. 2152–2156, Dec. 2021.

[53] J. Zhang, J. Wang, D. Li, J. Zhang, X. Sun, and X. Shang, "Hyperspectral band selection with dynamic graph enhancement and hierarchical feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5508816.

[54] L. Xu, M. Ahishali, and M. Gabbouj, "Dropout concrete autoencoder for band selection on hyperspectral image scenes," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[55] D. Yan, X. Sun, J. Zhang, and X. Shang, "Dual-view structural similarity subspace clustering for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[56] J. Wang et al., "Region-aware hierarchical latent feature representation learning-guided clustering for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 53, no. 8, pp. 5250–5263, Aug. 2023.

[57] C. Cui, X. Sun, B. Fu, and X. Shang, "SSANet-BS: Spectral–spatial cross-dimensional attention network for hyperspectral band selection," *Remote Sens.*, vol. 16, no. 15, p. 2848, Aug. 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/15/2848

[58] J. Jiang, J. Ma, Z. Wang, C. Chen, and X. Liu, "Hyperspectral image classification in the presence of noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 851–865, Feb. 2019.

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